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Effect of Hydrogen Enrichment on Methane–Air Combustion: Laminar Flame Speed, Ignition Delay and NO_x Emissions

Computer Methods in Combustion – Project Report

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Abstract

Blending hydrogen into natural-gas fuel streams is one of the short-term options considered for reducing the carbon intensity of existing gas infrastructure. However, hydrogen and methane differ substantially in their reactivity, diffusivity and adiabatic flame temperature, so even moderate H_2 addition can change the operability and emissions of a combustion system in non-obvious ways. In this project, the GRI-Mech 3.0 chemical kinetic mechanism is used together with the Cantera software package to study how the hydrogen molar fraction in a CH_4/H_2 fuel blend, $X_{H2} \in [0, 1]$, affects (i) the adiabatic flame temperature and equilibrium NO_x/CO emissions, (ii) the laminar flame speed S_L of freely propagating premixed flames, and (iii) the ignition delay time τ of a constant-volume adiabatic reactor. Results show that hydrogen enrichment increases the adiabatic flame temperature and equilibrium NO_x monotonically, increases the laminar flame speed by up to a factor of six at $X_{H2} = 1$, and reduces the ignition delay by one to two orders of magnitude at typical auto-ignition temperatures, with a qualitatively different (non-monotonic) pressure dependence for pure hydrogen compared to pure methane. The results are discussed in the context of practical hydrogen-blending limits for combustion equipment.

Contents

1 Introduction

Natural gas combustion remains one of the dominant sources of energy for power generation, industrial heat and domestic heating. In the transition towards lower-carbon energy systems, blending hydrogen into existing natural gas networks and burners has been proposed as a relatively low-cost intermediate step, since it can reuse a large part of the existing pipeline and combustion infrastructure [1]. The hydrogen fraction that can be tolerated by a given system, however, is limited by changes in the combustion characteristics of the fuel blend.

Hydrogen and methane differ strongly in their combustion-relevant properties:

- **Reactivity**: H_2 oxidation proceeds through the very fast $H + O_2 \longrightarrow OH + O$ branching reaction, giving H_2/air mixtures laminar flame speeds roughly six to eight times higher than CH_4/air mixtures, and ignition delay times that are orders of magnitude shorter at the same temperature.
- **Heating value**: on a mass basis, hydrogen has a lower heating value (LHV) of approximately 120 MJ/kg, about 2.4 times that of methane (50 MJ/kg), but on a volumetric basis hydrogen contains roughly three times less energy per unit volume.
- **Flame temperature and NO_x**: the higher adiabatic flame temperature of hydrogen-enriched mixtures tends to increase thermal NO_x formation via the Zeldovich mechanism [2].

The goal of this project is to quantify these effects for a CH_4/H_2/air system using detailed chemical kinetics, spanning the full blending range from pure methane ($X_{H2} = 0$) to pure hydrogen ($X_{H2} = 1$), where X_{H2} is the molar fraction of hydrogen in the fuel (i.e. the fuel composition is $(1 - X_{H2})\,CH_4 + X_{H2}\,H_2$). Specifically, three complementary numerical studies are carried out:

1. **Chemical equilibrium**: adiabatic flame temperature T_{ad} and equilibrium concentrations of NO, NO_2 and CO as a function of equivalence ratio ϕ and X_{H2}.
2. **Laminar flame speed**: one-dimensional freely-propagating premixed flame simulations giving $S_L(\phi, X_{H2})$ and its pressure dependence, as well as the flame temperature and species structure.
3. **Ignition delay**: zero-dimensional constant-volume adiabatic reactor simulations giving the auto-ignition delay time $\tau(T_0, p_0, X_{H2})$.

All simulations use the GRI-Mech 3.0 reaction mechanism [3] (53 species, 325 reactions) through the Cantera 3.2 Python interface [4].

2 Literature Review

The effect of hydrogen addition on the combustion properties of methane–air (and more generally natural-gas–air) mixtures has been studied extensively, both experimentally and numerically.

Di Sarli and Di Benedetto [5] measured and computed the laminar burning velocity and Markstein length of H_2/CH_4/air premixed flames over a wide range of hydrogen fractions and equivalence ratios. They found that the laminar burning velocity increases monotonically and increasingly steeply with the hydrogen content, with the strongest relative increase occurring above approximately 50% H_2 by volume in the fuel – a trend that is reproduced qualitatively by the present simulations (Section 4.2).

Hermanns et al. [6] carried out a systematic experimental and modelling study of the laminar burning velocity of $CH_4 + H_2 + O_2 + N_2$ flames as a function of unburned gas temperature, equivalence ratio and hydrogen fraction, providing one of the most complete datasets used for validation of kinetic mechanisms (including GRI-Mech) in this composition space.

Donohoe et al. [1] investigated ignition delay times and laminar flame speeds of natural-gas/hydrogen blends at engine-relevant elevated pressures, both experimentally (in a shock tube and rapid compression machine) and numerically. Their results show that hydrogen addition substantially reduces the ignition delay time, and that the pressure dependence of the ignition delay can differ qualitatively between methane-rich and hydrogen-rich mixtures because of the competition between the chain-branching reaction $H + O_2 \longrightarrow OH + O$ and the pressure-dependent chain-terminating reaction $H + O_2 + M \longrightarrow HO_2 + M$. This competition is discussed further in Section 4.3 in the context of the pressure scan results obtained here.

The GRI-Mech 3.0 mechanism [3] used in this work was developed and optimized primarily for natural-gas combustion (methane, with C_2 chemistry and NO_x sub-mechanism), but it also includes the core H_2/O_2 chemistry and is commonly used as a baseline mechanism for CH_4/H_2 blend studies, as in [5, 6, 1].

3 Model Description

3.1 Chemical kinetic mechanism and fuel definition

All simulations use the GRI-Mech 3.0 mechanism [3] (`gri30.yaml` in Cantera), which includes 53 species and 325 elementary reactions, and contains a NO_x sub-mechanism (thermal/Zeldovich, prompt and N_2O-intermediate routes) needed to predict NO and NO_2 formation.

The fuel is a binary blend of methane and hydrogen, parametrized by the hydrogen molar fraction in the fuel, $X_{H2} \in [0, 1]$:

$$\text{Fuel} = (1 - X_{H2})\, \text{CH}_4 + X_{H2}\, \text{H}_2 \tag{1}$$

so that $X_{H2} = 0$ corresponds to pure methane and $X_{H2} = 1$ to pure hydrogen. The oxidizer is air, represented as $O_2 : 1,\ N_2 : 3.76$ (molar). For a given equivalence ratio ϕ, Cantera's `set_equivalence_ratio` method is used to determine the fuel/air ratio relative to the stoichiometric ratio, defined in the usual way as

$$\phi = \frac{(n_{\text{fuel}}/n_{\text{O}_2})}{(n_{\text{fuel}}/n_{\text{O}_2})_{\text{stoich}}} \tag{2}$$

For the global reactions

$$\text{CH}_4 + 2\,\text{O}_2 \longrightarrow \text{CO}_2 + 2\,\text{H}_2\text{O} \tag{3}$$

$$\text{H}_2 + {}_1/{}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} \tag{4}$$

the stoichiometric oxygen requirement of the blend is $2(1 - X_{H2}) + \frac{1}{2}X_{H2}$ mol O_2 per mol fuel, which is computed automatically by Cantera from the mechanism's elemental composition. All simulations start from a fresh, unburned mixture at the specified initial temperature T_0 and pressure p_0.

The fuel-blend lower heating value (LHV), used to put the energy density of the blends into context, was computed from the molar enthalpies of the unburned fuel and the fully oxidized products (CO_2 and H_2O, with H_2O assumed gaseous, i.e. LHV) at 298 K:

$$\text{LHV} = \frac{n_{fuel}\,\bar{h}_{fuel} + n_{O_2}\,\bar{h}_{O_2} - n_{CO_2}\,\bar{h}_{CO_2} - n_{H_2O}\,\bar{h}_{H_2O}}{M_{fuel}} \tag{5}$$

where $\bar{h}$ are molar enthalpies obtained from the GRI-Mech thermodynamic data and M_{fuel} is the molar mass of the fuel blend.

3.2 Chemical equilibrium model (adiabatic flame temperature, NO_x/CO)

The adiabatic flame temperature and the equilibrium product composition are obtained by equilibrating the unburned mixture at constant enthalpy and pressure (`gas.equilibrate("HP")` in Cantera), which solves for the chemical equilibrium state reachable adiabatically from the given reactants:

$$H(T_{ad}, p, \mathbf{Y}_{eq}) = H(T_0, p, \mathbf{Y}_0) \tag{6}$$

where $\mathbf{Y}_0$ and $\mathbf{Y}_{eq}$ are the reactant and equilibrium product mass-fraction vectors, respectively. This gives the thermodynamic upper bound on the flame temperature and the equilibrium concentrations of all 53 species, including NO, NO_2 and CO, without requiring any information about reaction rates. All equilibrium calculations are performed at $T_0 = 300\,\mathrm{K}$, $p_0 = 1\,\mathrm{atm}$.

3.3 Laminar premixed flame model (1D)

The laminar flame speed and flame structure are computed with Cantera's `FreeFlame` class, which solves the steady, one-dimensional governing equations for a freely-propagating premixed flame in a coordinate system attached to the flame:

$$\text{Continuity:} \quad \dot{m} = \rho u = \text{const} \tag{7}$$

$$\text{Species:} \quad \dot{m}\frac{dY_k}{dx} = -\frac{d(\rho Y_k V_k)}{dx} + \dot{\omega}_k W_k \tag{8}$$

$$\text{Energy:} \quad \dot{m}c_p\frac{dT}{dx} = \frac{d}{dx}\left(\lambda\frac{dT}{dx}\right) - \sum_k \rho Y_k V_k c_{p,k}\frac{dT}{dx} - \sum_k \dot{\omega}_k h_k W_k \tag{9}$$

where $\dot{m}$ is the (eigenvalue) mass flux, Y_k the species mass fractions, V_k the diffusion velocities, $\dot{\omega}_k$ the net production rates and h_k the species enthalpies. The mass-flux eigenvalue, evaluated at the cold inlet, gives the laminar flame speed $S_L = \dot{m}/\rho_{unburned}$ (`flame.velocity[0]` in Cantera).

Mixture-averaged transport properties are used. The domain width is set to 0.03 m and an adaptive grid is used, with refinement controlled by `set_refine_criteria(ratio=3, slope=0.1, curve=0.2)`, which limits the maximum ratio between adjacent grid spacings and bounds the change in solution slope and curvature between grid points. The flame is solved with Cantera's damped Newton solver using the `auto=True` continuation strategy (`flame.solve(loglevel=0, auto=True)`), starting from an initial guess and progressively refining the mesh and relaxing transport/diffusion terms until convergence. All flame simulations use unburned conditions $T_0 = 300\,\mathrm{K}$ unless otherwise stated.

3.4 Ignition delay model (0D constant-volume reactor)

Auto-ignition is modelled using a homogeneous, constant-volume, adiabatic batch reactor (`IdealGasReactor` coupled to a `ReactorNet`). The reactor state evolves according to the species and energy conservation equations for a closed, constant-volume system:

$$\frac{dY_k}{dt} = \frac{\dot{\omega}_k W_k}{\rho} \tag{10}$$

$$\frac{dT}{dt} = -\frac{1}{\rho c_v}\sum_k u_k \dot{\omega}_k W_k \tag{11}$$

The reactor is initialized at the desired (T_0, p_0, ϕ, X_{H2}) state and integrated forward in time. The ignition delay time τ is defined as the time at which the temperature rise rate dT/dt reaches its maximum, i.e. the steepest part of the temperature trace, computed numerically as $\tau = \arg\max_t (dT/dt)$ using `numpy.gradient` on the stored (t, T) trace. A case is considered "non-igniting" within the simulated time window (and reported as not-a-number) if the total temperature rise is less than 400 K.

This single-stage definition of τ is appropriate for the present mixtures, which under the conditions studied show a single dominant thermal runaway. It would, however, not resolve two-stage ignition behaviour (a distinct low-temperature heat-release stage followed by the main ignition event), which can occur for larger hydrocarbons but is not significant for CH_4/H_2 blends in the temperature range studied here.

4 Results and Discussion

4.1 Adiabatic flame temperature and equilibrium emissions

Figure 1 shows the adiabatic flame temperature T_{ad} as a function of equivalence ratio ϕ for six hydrogen blend fractions $X_{H2} \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$. For every blend, T_{ad} peaks slightly on the rich side of stoichiometric ($\phi \approx$ 1.0–1.1), which is the well-known result of the balance between increasing heat release and the dissociation of CO_2/H_2O at very high temperatures. The peak temperature increases monotonically with X_{H2}:

Table 1: Maximum adiabatic flame temperature for each fuel blend.

X_{H2}	ϕ at $T_{ad,max}$	$T_{ad,max}$ [K]
0.0	1.02	2233.2
0.2	1.02	2243.3
0.4	1.05	2258.0
0.6	1.05	2280.5
0.8	1.05	2318.2
1.0	1.08	2396.3

Going from pure methane to pure hydrogen increases the maximum adiabatic flame temperature by about 163 K (2233.2 K $\rightarrow$ 2396.3 K), and the equivalence ratio of the peak shifts slightly toward the rich side as the hydrogen fraction increases. This is consistent with the higher adiabatic flame temperature of hydrogen-air flames reported in the literature [5, 6].

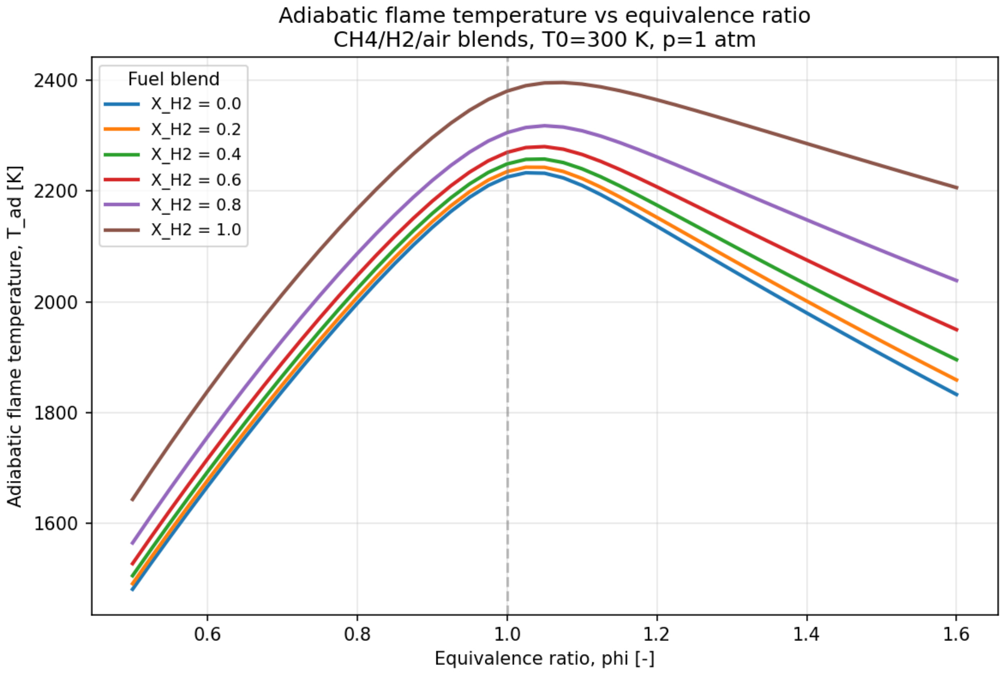

Figure 1: Adiabatic flame temperature vs equivalence ratio for CH_4/H_2/air blends ($T_0 = 300\,\mathrm{K}$, $p = 1\,\mathrm{atm}$).

Figure 2 shows the equilibrium $\mathrm{NO_x}$ ($NO+NO_2$) mole fraction versus ϕ. As expected from Zeldovich-mechanism thermal NO formation, $\mathrm{NO_x}$ peaks under slightly lean conditions ($\phi \approx$ 0.8–0.9), where the combination of high temperature and excess O_2 availability maximizes the thermal NO formation rate (equilibrium), and decreases sharply for rich mixtures where free O_2 is depleted. At $\phi = 1.0$, increasing X_{H2} from 0 to 1 increases equilibrium $\mathrm{NO_x}$ from 1888.6 ppm to 2529.5 ppm, an increase of about 34%, tracking the increase in T_{ad}.

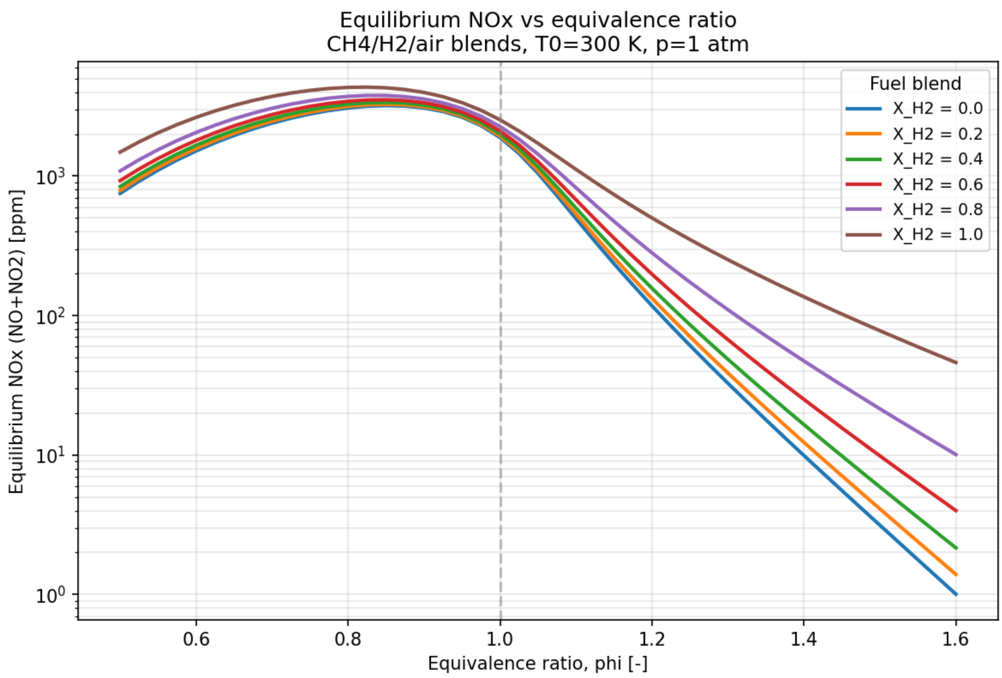

Figure 2: Equilibrium $\mathrm{NO_x}$ ($NO + NO_2$) vs equivalence ratio for CH_4/H_2/air blends.

Figure 3 shows the equilibrium CO mole fraction versus ϕ. CO is negligible for lean and near-stoichiometric mixtures but rises by orders of magnitude for rich mixtures ($\phi > 1$), as incomplete oxidation becomes thermodynamically favoured once the available O_2 is insufficient to fully oxidize the fuel carbon to CO_2. For a fixed ϕ, increasing X_{H2} *decreases* the equilibrium CO, simply because there is proportionally less carbon in the fuel blend; at $X_{H2} = 1$ (pure hydrogen) CO=0 by definition.

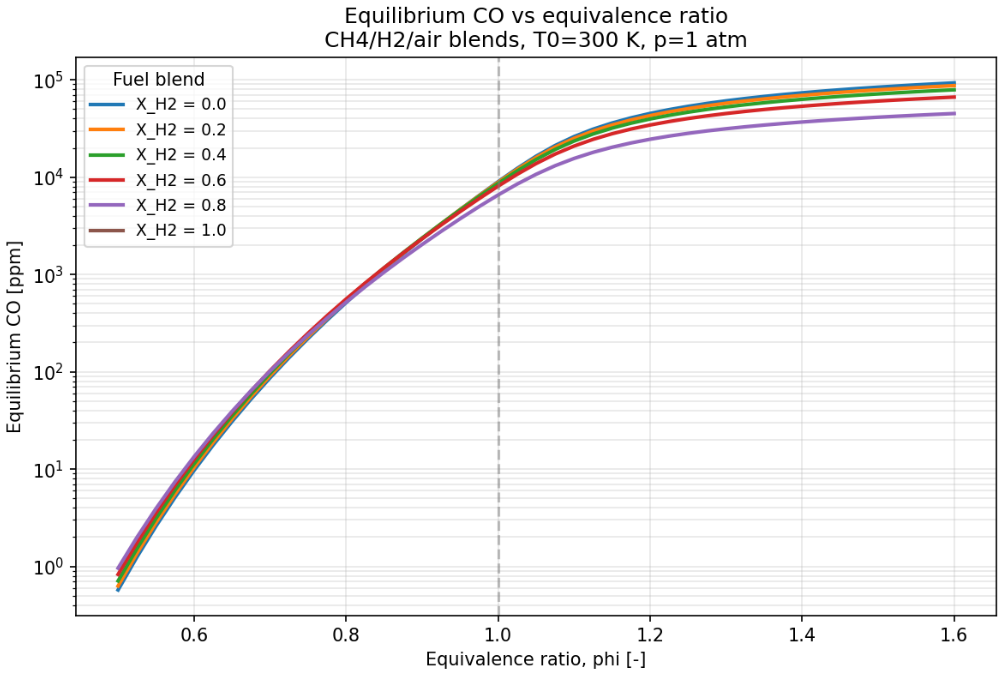

Figure 3: Equilibrium CO vs equivalence ratio for CH_4/H_2/air blends.

Figure 4 shows NO_x and CO as a function of X_{H2} for three equivalence ratios ($\phi = 0.8$, 1.0, 1.2). Several trends are visible:

- $\mathbf{NO_x}$ increases monotonically with X_{H2} at all three equivalence ratios. The lean mixture ($\phi = 0.8$) has the highest absolute NO_x levels (1888.6 ppm to 4322.8 ppm as X_{H2} goes from 0 to 1) – higher than at $\phi = 1.0$ – despite having a *lower* flame temperature (e.g. $T_{ad} = 1996.9\,\mathrm{K}$ at $\phi = 0.8, X_{H2} = 0$ vs $T_{ad} = 2225.5\,\mathrm{K}$ at $\phi = 1.0, X_{H2} = 0$). This apparent contradiction is explained by the equilibrium O_2 availability: thermal NO formation ($N_2 + O \rightleftharpoons NO + N$, $N + O_2 \rightleftharpoons NO + O$) requires free O and O_2, which are abundant under lean conditions but nearly absent at $\phi = 1.2$. The relative NO_x increase with X_{H2} is largest for the rich mixture ($\phi = 1.2$: from 117.4 ppm to 500.7 ppm, a factor of ∼4.3), because the higher H/O ratio of the hydrogen-rich blend leaves more residual O_2/O available even under globally fuel-rich conditions.

- **CO** decreases monotonically with X_{H2} for $\phi = 1.0$ and $\phi = 1.2$ (reaching exactly zero at $X_{H2} = 1$), as discussed above. At $\phi = 0.8$, CO is non-monotonic, rising slightly with X_{H2} up to about $X_{H2} \approx 0.6$ (from 514.2 ppm to 559.3 ppm) before falling sharply to zero – a consequence of the higher flame temperature at larger X_{H2} promoting CO_2 dissociation back to CO, until the diminishing amount of carbon in the blend dominates.

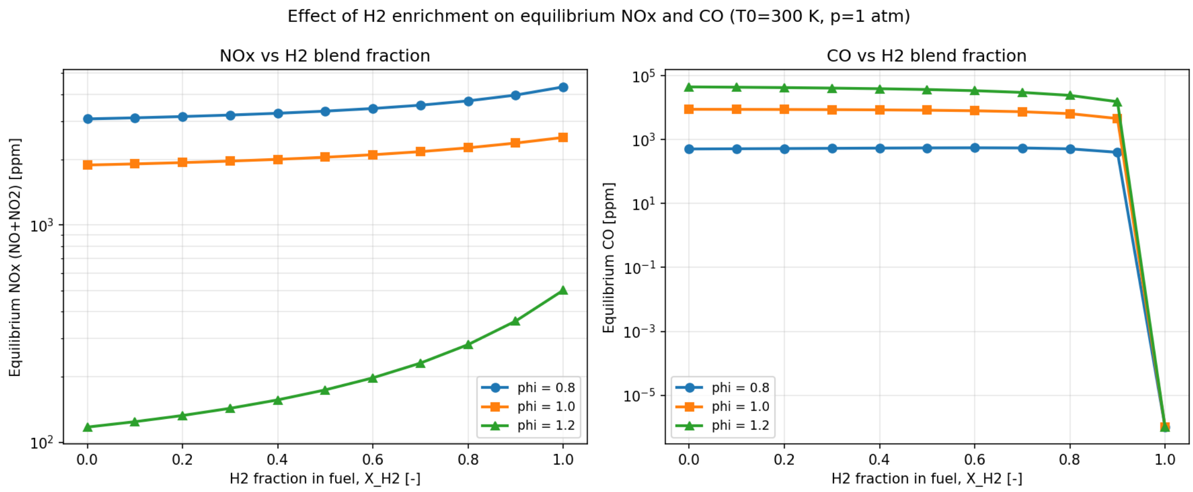

Figure 4: Equilibrium NO_x and CO vs hydrogen blend fraction X_{H2}, for $\phi = 0.8$, 1.0 and 1.2.

4.2 Laminar flame speed

Figure 5 shows the laminar flame speed S_L as a function of equivalence ratio for the five fuel blends $X_{H2} \in \{0, 0.25, 0.5, 0.75, 1.0\}$. For all blends, S_L peaks on the slightly rich side of stoichiometric ($\phi \approx 1.0$–1.15), consistent with the well-known behaviour of hydrocarbon and hydrogen flames. The peak flame speed increases strongly and increasingly steeply with X_{H2}:

Table 2: Laminar flame speed S_L [cm/s] vs equivalence ratio and hydrogen fraction ($T_0 = 300\,\mathrm{K}$, $p = 1\,\mathrm{atm}$).

X_{H2}	$\phi = 0.70$	$\phi = 0.85$	$\phi = 1.00$	$\phi = 1.15$	$\phi = 1.30$
0.00	19.6	31.3	38.5	37.1	23.9
0.25	23.4	37.3	46.1	45.7	32.4
0.50	30.6	48.3	60.7	63.1	50.6
0.75	47.7	75.3	97.9	108.8	102.2
1.00	122.0	185.2	234.0	270.0	294.2

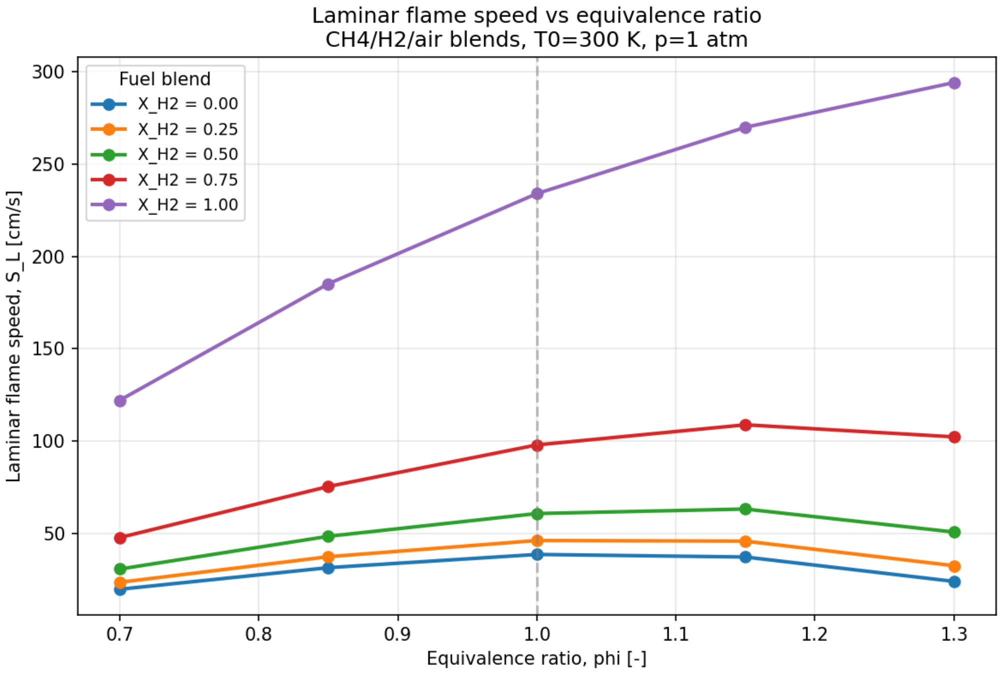

Figure 5: Laminar flame speed vs equivalence ratio for CH_4/H_2/air blends.

The pure-methane value $S_L \approx 38.5\,\mathrm{cm/s}$ at $\phi = 1.0$ agrees well with the commonly quoted experimental range of 36 cm/s to 40 cm/s for methane–air flames at atmospheric conditions. Two effects are particularly notable:

- The flame speed increase with X_{H2} is strongly *nonlinear*: at $\phi = 1.0$, going from $X_{H2} = 0$ to 0.5 roughly increases S_L from 38.5 to 60.7 cm/s (a 1.6-fold increase), while going from $X_{H2} = 0.5$ to 1.0 increases S_L from 60.7 to 234.0 cm/s (a 3.9-fold increase) – an overall increase of about a factor of 6 from pure methane to pure hydrogen. This is shown explicitly in Figure 6, where the steepest rise occurs above $X_{H2} \approx 0.5$. The dominant cause is the very high diffusivity of H_2 and the increasing role of the chain-branching reaction $H + O_2 \longrightarrow OH + O$, whose rate increases superlinearly as H_2 becomes a larger fraction of the fuel.

- The location of the peak S_L shifts toward richer mixtures as X_{H2} increases: for $X_{H2} = 0$ the peak is near $\phi \approx 1.0$, whereas for $X_{H2} = 1$ the flame speed is still increasing at $\phi = 1.30$ (the richest case considered), consistent with the well-known shift of the peak burning velocity of hydrogen–air flames to $\phi \approx 1.6$–1.8 due to preferential diffusion effects.

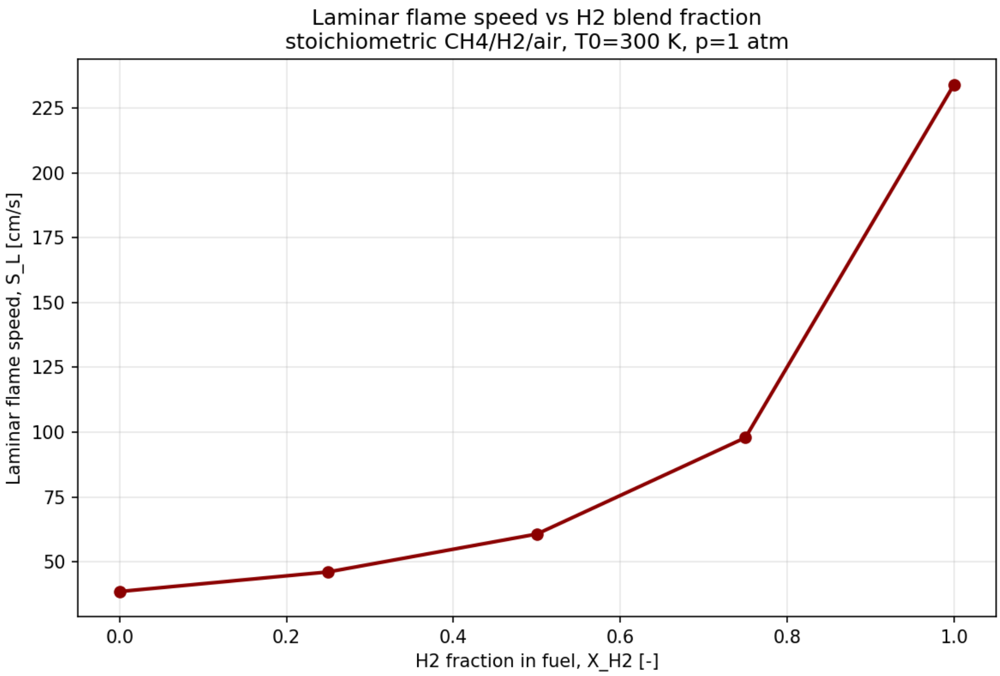

Figure 6: Laminar flame speed vs hydrogen blend fraction X_{H2} at stoichiometric conditions ($\phi = 1.0$).

Figure 7 shows the effect of pressure on S_L at stoichiometric conditions for $X_{H2} \in \{0, 0.5, 1.0\}$, over $p \in \{1, 3, 6\}$ bar:

Table 3: Laminar flame speed S_L [cm/s] vs pressure at $\phi = 1.0$, $T_0 = 300\,\text{K}$.

X_{H2}	$p = 1\,\text{bar}$	$p = 3\,\text{bar}$	$p = 6\,\text{bar}$
0.00	38.7	24.8	17.7
0.50	61.0	39.6	28.5
1.00	234.0	215.8	181.7

For all three blends, S_L decreases with increasing pressure, in agreement with the classical $S_L \propto p^{n-1}$ scaling for an overall reaction order $n < 1$ (typically $n \approx 1.6$–1.8 for hydrocarbon and hydrogen flames, giving $n - 1 < 0$). However, the *relative* sensitivity to pressure differs markedly between the blends: from 1 to 6 bar, S_L for pure methane drops by about 54% ($38.7 \to 17.7$ cm/s), whereas for pure hydrogen it drops by only about 22% ($234.0 \to 181.7$ cm/s). This is consistent with the higher overall reaction order (closer to 2) typically reported for hydrogen flames compared with methane flames, making hydrogen flame speed comparatively more robust to pressure increase – a relevant consideration for pressurized combustion systems (e.g. gas turbines) operating on hydrogen-enriched fuel.

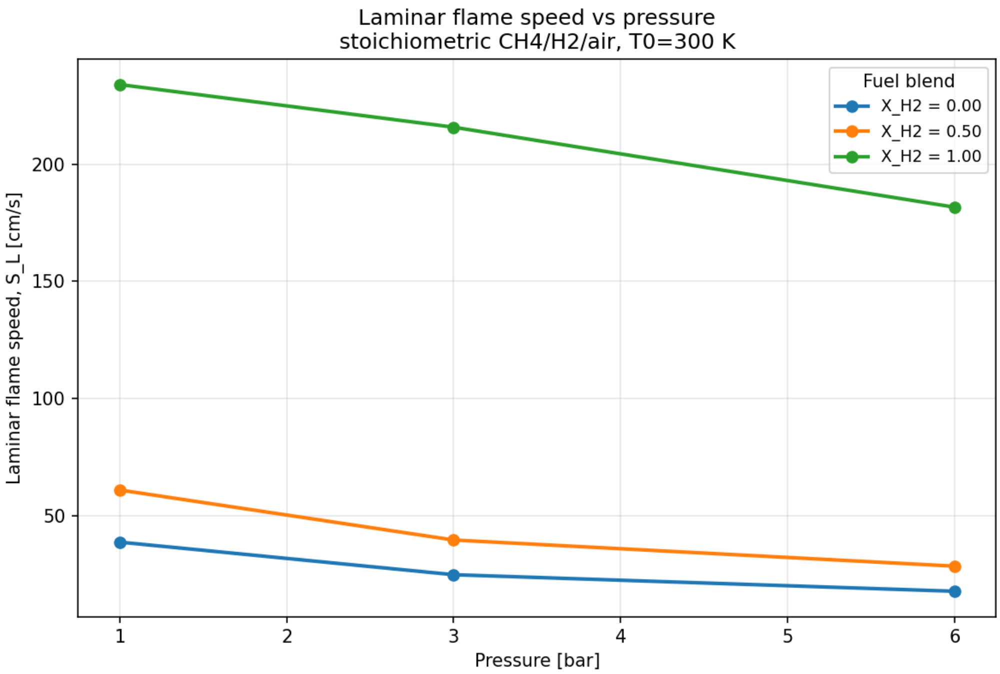

Figure 7: Laminar flame speed vs pressure at $\phi = 1.0$ for $X_{H2} = 0$, 0.5 and 1.0.

Finally, Figure 8 shows the spatial structure (temperature and major species mole fractions) of the stoichiometric $X_{H2} = 0.5$ flame. The computed burned-gas temperature ($T \approx 2260\,\mathrm{K}$) matches the equilibrium adiabatic flame temperature for the same composition (2258.4 K, Section 4.1) to within 0.1%, confirming the consistency between the equilibrium and 1D flame models. The reaction zone is only a fraction of a millimetre thick, with OH – a marker of the high-temperature reaction zone – peaking sharply at the location where CH_4 and H_2 are simultaneously consumed.

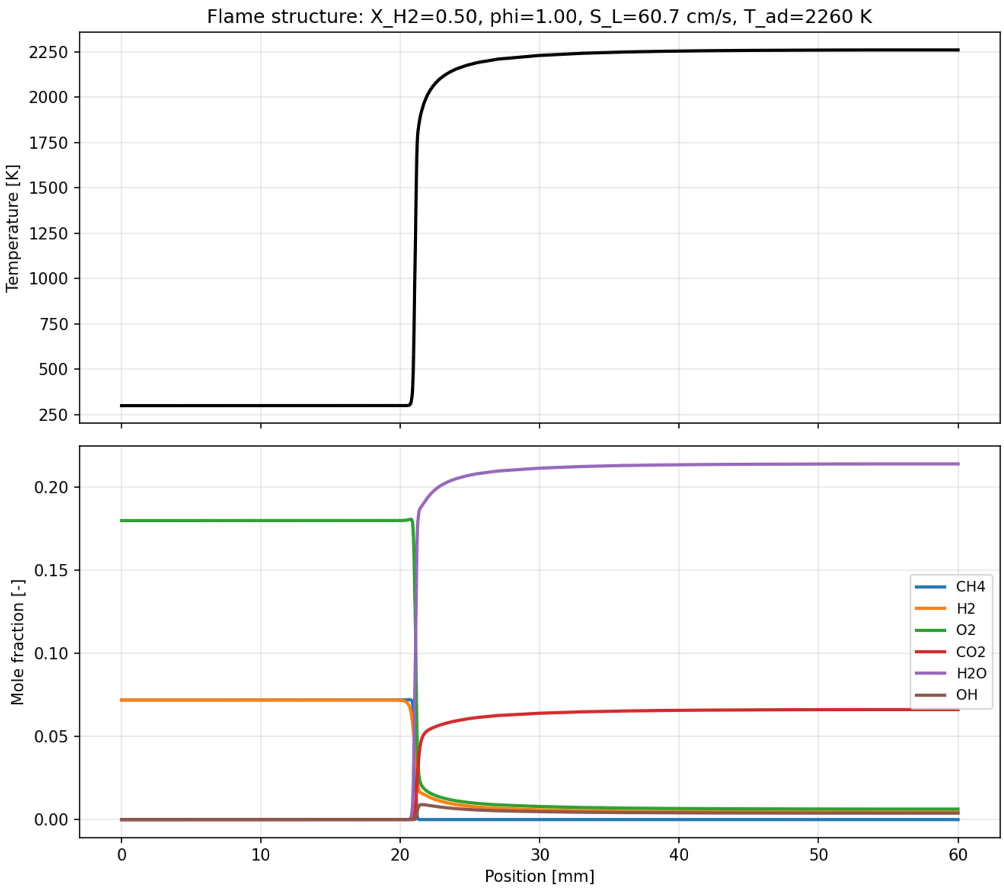

Figure 8: Flame structure (temperature and major species mole fractions) for the stoichiometric $X_{H2} = 0.5$ flame.

4.3 Ignition delay

Figure 9 shows the ignition delay time τ as a function of $1000/T_0$ (an Arrhenius-type representation) at $p_0 = 10$ bar, $\phi = 1.0$, for the five fuel blends. As expected, τ decreases approximately exponentially with increasing T_0 for all blends (an approximately straight line on the semi-logarithmic plot), reflecting the Arrhenius temperature dependence of the dominant chain-branching reaction rate. At every temperature, increasing X_{H2} reduces τ, and the effect is dramatic:

Table 4: Ignition delay time τ [ms] vs initial temperature T_0 and hydrogen fraction ($p_0 = 10$ bar, $\phi = 1.0$).

T_0 [K]	$X_{H2} = 0.00$	$X_{H2} = 0.25$	$X_{H2} = 0.50$	$X_{H2} = 0.75$	$X_{H2} = 1.00$
900	594.4	165.7	158.6	137.3	101.8
1000	80.9	12.85	11.49	10.20	8.106
1100	16.98	1.737	1.154	0.9807	0.8342
1200	4.528	0.4906	0.1954	0.1036	0.06067
1400	0.4721	0.1027	0.03332	0.01066	0.002284

At $T_0 = 1100$ K, adding just 25% H_2 (by mole, in the fuel) already reduces τ from 16.98 ms to 1.737 ms – nearly a 10-fold reduction – while the further reduction from $X_{H2} = 0.25$ to $X_{H2} = 1.0$ (1.737 → 0.834 ms) is comparatively modest. The same pattern (a very steep drop from $X_{H2} = 0$

to ≈ 0.25, followed by a much shallower decrease) is visible across the whole temperature range and is shown explicitly in Figure 10. This indicates that even a relatively small admixture of hydrogen into methane has a disproportionately large effect on the auto-ignition propensity of the blend – an important safety consideration for hydrogen-blending in gas networks and combustion systems originally designed for pure methane.

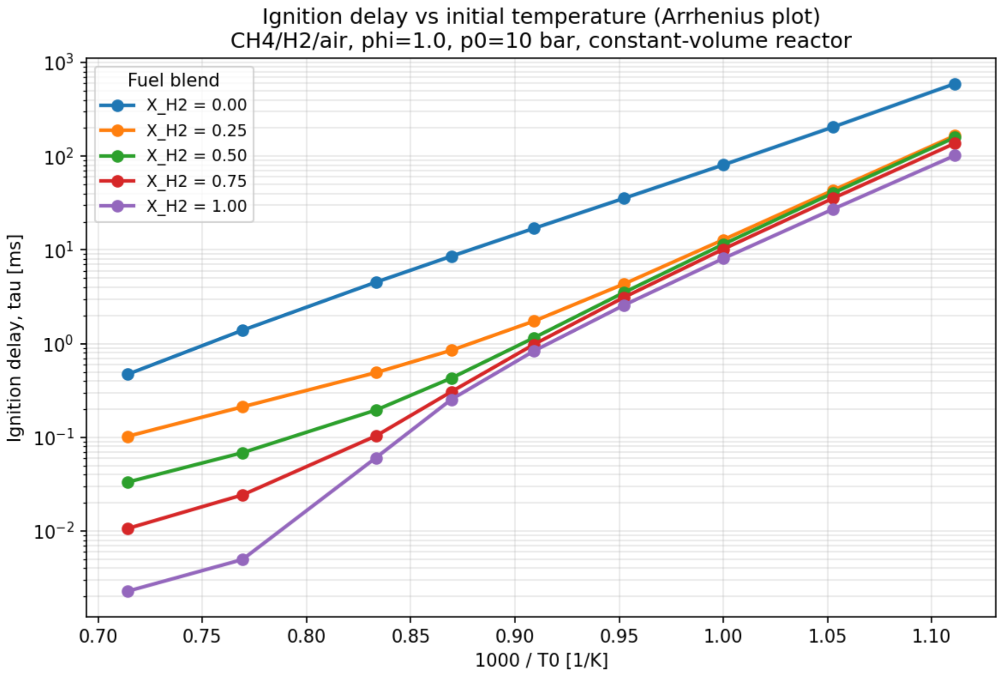

Figure 9: Ignition delay time vs $1000/T_0$ (Arrhenius plot) at $p_0 = 10\,\text{bar}$, $\phi = 1.0$.

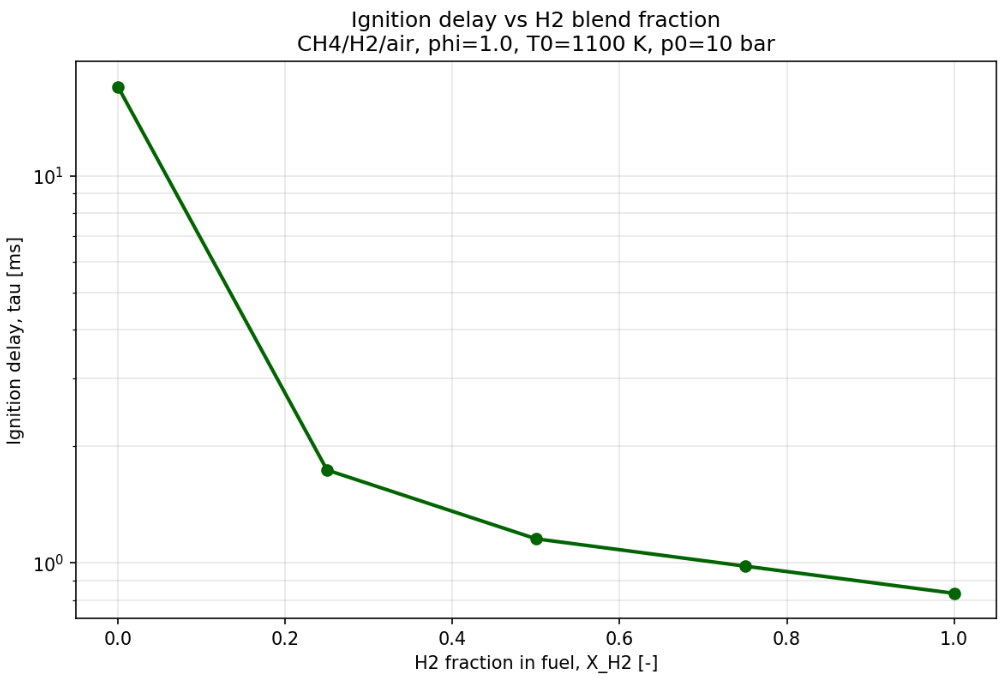

Figure 10: Ignition delay time vs hydrogen blend fraction X_{H2} at $T_0 = 1100\,\text{K}$, $p_0 = 10\,\text{bar}$, $\phi = 1.0$.

Figure 11 and Table 5 show the pressure dependence of τ at $T_0 = 1100\,\mathrm{K}$ for $X_{H2} \in \{0, 0.5, 1.0\}$:

Table 5: Ignition delay time τ [ms] vs initial pressure p_0 at $T_0 = 1100\,\mathrm{K}$, $\phi = 1.0$.

X_{H2}	$p_0 = 1\,\mathrm{bar}$	$p_0 = 10\,\mathrm{bar}$	$p_0 = 25\,\mathrm{bar}$
0.00	200.4	16.98	6.101
0.50	2.001	1.154	0.9594
1.00	0.08702	0.8342	0.6643

For pure methane ($X_{H2} = 0$), τ decreases monotonically with pressure ($200.4 \to 16.98 \to 6.101\,\mathrm{ms}$ from 1 to 25 bar), as expected for a chain-branching system where increased pressure increases the collision/ reaction rates. However, for pure hydrogen ($X_{H2} = 1$), the trend is qualitatively *opposite* at low pressure: τ *increases* by almost an order of magnitude from 1 to 10 bar ($0.087 \to 0.834\,\mathrm{ms}$) before decreasing again from 10 to 25 bar ($0.834 \to 0.664\,\mathrm{ms}$). The $X_{H2} = 0.5$ blend shows an intermediate, much weaker version of the same non-monotonic behaviour ($2.00 \to 1.15 \to 0.959\,\mathrm{ms}$).

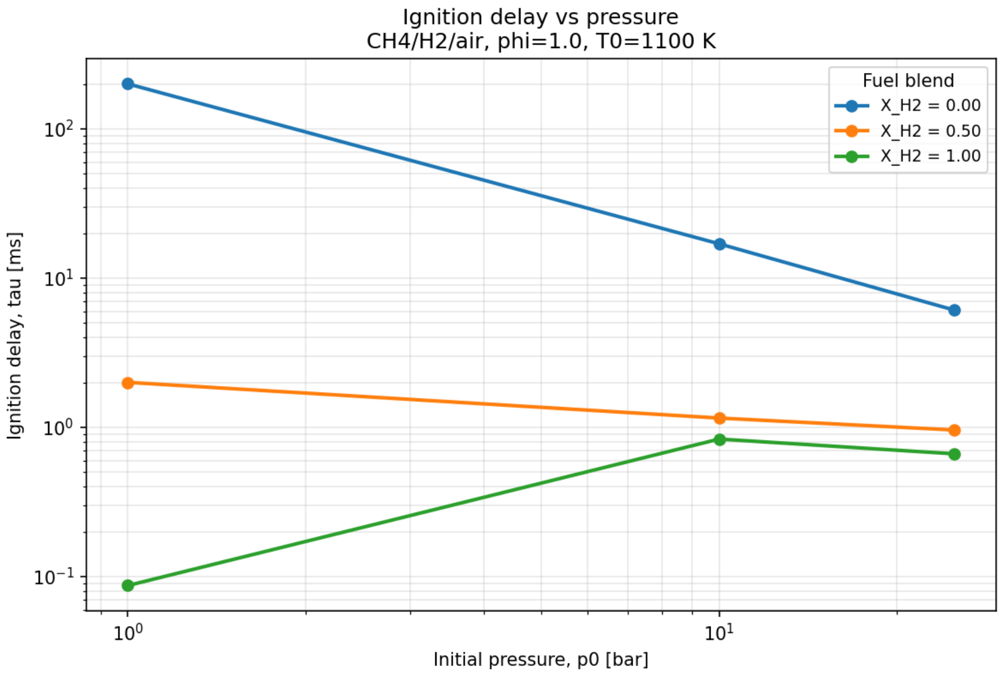

Figure 11: Ignition delay time vs pressure at $T_0 = 1100\,\mathrm{K}$, $\phi = 1.0$, for $X_{H2} = 0$, 0.5 and 1.0.

This non-monotonic behaviour for hydrogen-rich mixtures is a well-documented consequence of the competition between two reactions involving the H radical [1]:

$$\mathrm{H} + \mathrm{O_2} \longrightarrow \mathrm{OH} + \mathrm{O} \qquad \text{(chain branching, pressure-independent)} \tag{12}$$

$$\mathrm{H} + \mathrm{O_2} + \mathrm{M} \longrightarrow \mathrm{HO_2} + \mathrm{M} \qquad \text{(chain terminating, third-order, } \propto p^2\text{)} \tag{13}$$

The first reaction produces two reactive radicals (OH and O) from one H atom and drives the chain reaction forward; the second consumes an H radical to form the comparatively unreactive $\mathrm{HO_2}$, removing radicals from the branching chain. Because the second reaction is third-order, its rate grows with the square of the pressure, while the first reaction's rate is pressure-independent. At low pressure, branching dominates and increasing pressure increases the overall reaction rate (shorter τ); but for fuels whose ignition chemistry is dominated by the H/O_2 system (i.e. hydrogen-rich blends), there is a pressure range where the growth of the termination reaction outpaces the branching reaction, causing the net radical pool to grow more slowly and τ to *increase* with pressure – the "ignition delay turnover" behaviour seen here for $X_{H2} = 1$ between 1 and 10 bar. For methane, the larger relative contribution of CH_4-specific branching pathways (which do not involve this particular three-body step in the same way) means the overall ignition delay remains monotonically decreasing with pressure over the range studied.

5 Conclusions

A systematic numerical study of CH_4/H_2/air combustion was carried out using the GRI-Mech 3.0 mechanism in Cantera, covering chemical equilibrium, one-dimensional laminar premixed flames, and zero-dimensional auto-ignition, for hydrogen blend fractions X_{H2} from 0 (pure methane) to 1 (pure hydrogen). The main findings are:

1. **Adiabatic flame temperature** increases monotonically with X_{H2}, from a maximum of 2233.2 K (pure CH_4, $\phi \approx 1.02$) to 2396.3 K (pure H_2, $\phi \approx 1.08$), with the peak shifting slightly toward richer mixtures as X_{H2} increases.
2. **Equilibrium NO_x** increases monotonically with X_{H2} at all equivalence ratios studied (e.g. by ∼34% at $\phi = 1.0$, from 1888.6 to 2529.5 ppm), tracking the increase in flame temperature. Counter-intuitively, lean mixtures ($\phi = 0.8$) show *higher* equilibrium NO_x than stoichiometric mixtures despite lower flame temperatures, because of the much higher residual O_2/O concentration available to drive thermal NO formation.
3. **Equilibrium CO** decreases (toward zero) with increasing X_{H2} at stoichiometric and rich conditions, simply reflecting the reduced carbon content of the fuel blend, but shows a slight non-monotonic increase at lean conditions before falling to zero.
4. **Laminar flame speed** increases strongly and nonlinearly with X_{H2} – by a factor of ∼6 at stoichiometric conditions, from 38.5 cm/s (pure CH_4) to 234.0 cm/s (pure H_2) – with the steepest increase occurring for $X_{H2} \gtrsim 0.5$. The location of peak S_L shifts toward richer mixtures with increasing X_{H2}.
5. **Pressure sensitivity of S_L** is markedly different between fuels: from 1 to 6 bar, pure methane's flame speed drops by ∼54% while pure hydrogen's drops by only ∼22%, making hydrogen-enriched flames comparatively more robust at elevated pressure.
6. **Ignition delay** decreases dramatically with X_{H2}, with the largest relative effect occurring already for small hydrogen additions ($X_{H2} = 0 \rightarrow 0.25$ gives nearly a 10-fold reduction at $T_0 = 1100$ K, $p_0 = 10$ bar); further increases in X_{H2} have a comparatively smaller effect.
7. **Pressure dependence of ignition delay** is qualitatively different between fuels: monotonically decreasing for pure methane, but non-monotonic (increasing then decreasing) for pure hydrogen, due to the competition between the pressure-independent chain-branching reaction $H{+}O2 \longrightarrow OH{+}O$ and the third-order, pressure-squared chain-terminating reaction $H{+}O2{+}M \longrightarrow HO2{+}M$.

Practical implications

These results illustrate that even moderate hydrogen blending into methane-fuelled systems can have disproportionately large effects on key combustion characteristics – particularly flame speed (affecting flashback/blow-off limits of burners) and ignition delay (affecting auto-ignition safety margins and pre-ignition risk in engines) – well before X_{H2} approaches 1. NO_x emissions also increase steadily with hydrogen content, which may partially offset the CO_2 reduction benefits of hydrogen blending unless burner design is adapted (e.g. via leaner operation or staged combustion).

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